

# Karan Shah

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## Education

### SUNY, Stony Brook University

MS IN COMPUTER SCIENCE | GPA : 3.64/4

Coursework: Data Science, Probability and Statistics, Machine Learning, Big Data Analytics, Computer Vision, Theory of Database Systems

Stony Brook, NY

Aug. 2019 - Present

### Gujarat Technological University

B.E. IN COMPUTER ENGINEERING

Coursework : Data Structures, Algorithms, Software Engineering, Object Oriented Programming, Distributed Operating System

Ahmedabad, India

Aug. 2014 - May. 2018

## Skills

**IDEs** Visual Studio, NetBeans, Jupyter Notebook, Eclipse, IntelliJ IDEA, Vim, Sublime

**Programming** Python, Javascript, JAVA, C/C++, Golang

**Tools/Technologies** Matlab, AWS, MapReduce, Spark, Hadoop, Git, SQL, NoSQL, Linux, Docker, Jenkins, Kubernetes

**Frameworks/Libraries** Scikit-Learn, Numpy, Pandas, PyTorch, Tensorflow, Keras, OpenCV, NodeJS, Django, Flask, Angular, ElectronJS

## Work Experience

### SUNY, Stony Brook University

GRADUATE TEACHING ASSISTANT

• Assisted the professor in Course CSE327-01: **Computer Vision** for assignments formulations, assessments and resolving queries of students

Stony Brook, NY

January 2020 - May 2020

### Simform Solutions

SOFTWARE ENGINEER

- Developed and automated ETL jobs to work on a huge realtime data using AWS services (Glue, Lambda, S3) and Spark
- Developed an end to end Video Streaming system; able to upload, scan and host the content on the cloud, and stream on cross-platform desktop application over globally separated clients; using AWS services (Lambda, S3, Cloudfront, MQTT), NodeJS, and Python
- Developed software system to semi-automate Company's marketing work with Chrome Extension and REST API Web backend using Python and NodeJS, efficient in reducing ~ **40 %** of Marketing team's manual work

Ahmedabad, India

June 2018 - April 2019

### ISRO - Indian Space Research Organization

RESEARCH INTERN, MACHINE LEARNING

- Designed generalized **Convolutional Neural Network** architecture to handle multimodal and/or multispectral imagery along with adaptability to varying application requirements; trained on Potsdam benchmark dataset producing State of the art results to **Semantic Segmentation**
- Built software tools in **Python** for data preparation/processing of geographic remote sensing data, visualization, real-time training performance evaluation, post-processing and output generation as a semi-automation approach
- Manipulated raster GIS data as a part of the pre-processing task using Python and QGIS application

Ahmedabad, India

July 2017 - May 2018

## Publication

### Band-wise Independent Pansharpening Using Convolutional Neural Networks with Shared Weights

PANKAJ BODANI, KARAN SHAH, SHASHIKANT SHARMA

- Submitted to **IEEE** Geoscience and Remote Sensing Letters

## Projects

### Child Mortality Analysis, Under Supervision of Prof. Andrew Schwartz ( Python, Dask, Spark )

April 2020 - May 2020

- Analysed secondary factors contributing to child mortality through Big Data Techniques
- Implemented Linear Regression and Similarity Search running on women lifestyle real data distributed on Hadoop clusters

### Covid Data Analysis, Instructed by Prof. Anshul Gandhi ( Python, Pandas, Numpy )

March 2020 - April 2020

- Analysed Covid's impact using a Covid dataset and an associated (non-covid) dataset
- Performed AR, EWMA, Wald's test, Z-test, t-test, K-S test, permutation test, correlation, and Bayesian inference along with suitable visualizations

### Retail Sales Data Analysis, Under Supervision of Prof. Steven Skiena ( Python, Pandas )

Oct. 2019 - Dec. 2019

- Worked on a market basket dataset of a local chain retail stores to provide insights to help the company with their business
- Analysed data to find reasons for drop in sales with additional help of external datasets
- Implemented **Word2Vec** and **LSTM model with k-means clustering** to improve Shelf Space Management and Inventory management

### Pansharpening Using Convolutional Neural Networks ( Python, Keras )

Jan. 2019 - July 2019

- Fine-tuned a Convolutional Neural Network to super-resolve Multispectral Satellite Imagery to Panchromatic Resolution
- Delivered high performance in terms of reduced resolution quantitative assessment and visual inspection
- Achieved **better results** compared to popular methods used for this purpose in the industry

### Video Action Classification and Recognition ( Python, PyTorch )

Nov. 2019 - Dec. 2019

- Trained CNN for human action recognition task on the UCF101 data
- Achieved **86%** (inclass 3rd rank) on image test data, and **84%** (inclass 4th rank) on video frames test data
- Trained LSTM in RNN to actions classification on data collected by Kinect v2
- Used Transfer Learning to compute features for 60000 video frames with limited compute resources.

### Ultrasound Nerve Segmentation (Python, Tensorflow, Keras)

Nov. 2017 - Jan. 2018

- Designed and trained CNN based on U-net to segment a collection of Nerves in Ultrasound Images taken from a Kaggle competition
- Used pre-trained weights of VGG model in the Encoder of the network (transfer learning), gained **68 %** Test Accuracy (Top 100)